

ADVANCED POWER BI PRACTICAL QUESTION PAPER

Course Evaluation • ICA / MCA Data Analytics Practice Examination

Exam Objective & Scope: Complete exam practice paper covering end-to-end business intelligence workflow: Data Modeling (Star Schema), Power Query ETL, Complex DAX, KPIs, Visualizations, Matrix Reporting, Slicers & Filters, Drill-Through, Bookmarks, Custom Tooltips, and Multi-page Dashboard Deployment.

Core Dataset: Sales Performance Dashboard • Sales Products Customers Employees Region Date Table

SECTION 1 — DATA IMPORT & POWER QUERY ETL

Perform the following data transformation tasks within Power Query Editor prior to loading the model:

- **Data Extraction:** Import all required transactional and dimensional tables from Excel workbook.
- **Schema & Data Types:** Inspect and enforce correct data types (Dates, Decimals, Integers, Text) across all fields.
- **Data Hygiene:** Identify and eliminate duplicate records; handle and replace null/blank values systematically.
- **Text Transformations:** Split Customer Name into separate columns: First Name and Last Name .
- **Address Synthesis:** Create a consolidated column Full Address by concatenating City and State .
- **Column Governance:** Standardize and rename all column headers to follow clean PascalCase / Title Case naming conventions.
- **Custom Metrics:** Create a custom column computing initial row-level Profit .
- **Load Workflow:** Execute Close & Apply to commit staging transformations to the data model.

SECTION 2 — DATA MODELING (STAR SCHEMA)

Design and configure a robust Star Schema with 1-to-many (1:*) single-direction relationships:

Fact Table (Many side)	Dimension Table (One side)	Key Field
Sales	Products	ProductID
Sales	Customers	CustomerID
Sales	Employees	EmployeeID
Sales	Region	RegionID
Sales	Date Table	Date

DATE DIMENSION BUILD REQUIREMENTS:

- Generate an autonomous continuous Date Table using DAX: CALENDAR() or CALENDARAUTO() .
- Derive calculated fields: Year , Quarter , Month , Month Number , Week Number , and Day Name .
- Configure column sorting hierarchy: Sort Month Name strictly by Month Number to avoid alphabetical ordering.
- Explicitly configure: Mark as Date Table referencing the primary Date field.

SECTION 3 — CALCULATED COLUMNS (DAX)

Create the following calculated columns using **IF**, **SWITCH**, **YEAR**, **MONTH**, and **FORMAT**:

Column Name	DAX Logic & Business Requirement
Profit	Compute gross margin: SalesAmount - CostAmount
Sales Status	Categorize transaction volume into High / Medium / Low thresholds using nested IF() or SWITCH(TRUE(), ...)
Profit Status	Classify bottom-line performance as Profit or Loss based on net value
Region Category	Group geographic zones into North India / South India / East India / West India
Month Name	Extract descriptive month text from Date utilizing FORMAT()
Quarter	Derive fiscal quarter designation formatted as Q1 – Q4

SECTION 4 — CORE DAX MEASURES SPECIFICATION

Implement robust, optimized DAX measures grouped logically within a dedicated Measures Table:

1. CORE SALES MEASURES

- Total Sales** = **SUM(Sales[SalesAmount])**
- Total Cost** = **SUM(Sales[CostAmount])**
- Total Profit** = **[Total Sales] - [Total Cost]**
- Total Quantity** = **SUM(Sales[Quantity])**
- Total Orders** = **DISTINCTCOUNT(Sales[OrderID])**
- Average Order Value** = **DIVIDE([Total Sales], [Total Orders], 0)**
- Average Profit** = **AVERAGE(...)** or order-level average

2. TIME INTELLIGENCE MEASURES

- Previous Month Sales** (via **DATEADD**)
- MoM Sales Change** & **MoM Growth %**
- Previous Year Sales** (via **SAMEPERIODLASTYEAR**)
- YoY Growth %** (using **DIVIDE**)
- YTD Sales** & **YTD Profit** (via **TOTALYTD**)
- MTD Sales** & **QTD Sales** (via **TOTALMTD** / **TOTALQTD**)

3. RANKING & ITERATION MEASURES

- Rank Product by Sales** — dynamic rank using **RANKX(ALL(Products), [Total Sales], ...)**
- Rank Region by Profit** — ranking across region dimension
- Top 5 Products Sales** — computed using **CALCULATE** and **TOPN(5, ...)**

4. TARGET & VARIANCE MEASURES

- Sales Target** & **Profit Target**
- Sales Variance** = **[Total Sales] - [Sales Target]**
- Target Achievement %** = **DIVIDE([Total Sales], [Sales Target], 0)**
- Profit Achievement %** = **DIVIDE([Total Profit], [Profit Target], 0)**

REQUIRED DAX FUNCTIONS:

CALCULATE	DATEADD	SAMEPERIODLASTYEAR
TOTALYTD	TOTALMTD	TOTALQTD
	DIVIDE	RANKX
TOPN		

SECTION 5 — KPI DASHBOARD CONSTRUCTION

Build an Executive KPI summary row leveraging Card visuals and the native Power BI **KPI Visual**:

- Executive Metric Cards:** Display **Total Sales**, **Total Profit**, **Profit Margin %**, **Total Orders**, **Average Order Value**, **MoM Growth %**, **YoY Growth %**, and **Target Achievement %**.
- Standard KPI Visual:**
 - Indicator:** **Total Sales**
 - Trend Axis:** **Date Table[Date]** (Monthly hierarchy)
 - Target Goals:** **Sales Target**

SECTION 6 — VISUALIZATIONS PORTFOLIO

Construct all twelve specified analytical visuals formatted with appropriate data labels, titles, and legends:

Visual Type	Analytical Requirement & Field Mappings
Clustered Column Chart	Sales distribution categorized by Region
Stacked Column Chart	Sales segmented by Product Category across geographical Region
Bar Chart	Horizontal comparison of Profit sorted across individual Product items
Line Chart	Continuous time-series analysis representing Monthly Sales Trend
Pie Chart	Composition of total Sales broken down by Customer Segment
Donut Chart	Share of Sales contribution by Product Category
Scatter Chart	Correlation analysis: X-Axis = Sales , Y-Axis = Profit , Details = Product
Funnel Chart	Lead and sales conversion velocity mapped by sales pipeline Stage
Treemap	Hierarchical area-based volume mapping of Sales by Product Category
Combo (Line & Column)	Actual Sales (Columns) tracked against projected Sales Target (Line)
Waterfall Chart	Variance decomposition showing monthly incremental Profit increase/decrease
Ribbon Chart	Dynamic rank transitions of Region performance tracked over chronological periods

SECTION 7 — MATRIX VISUAL (PIVOT TABLE STYLE REPORTING)

Configure an advanced multi-tiered financial matrix supporting full drill-down interaction:

- **Row Hierarchy:** **Year** → **Quarter** → **Month** → **Product Category**
- **Column Grouping:** **Region**
- **Values:** **Sales** , **Profit** , **Quantity** , **Profit Margin %**
- **Drill Configuration:** Enable **Drill Down** and **Expand All** navigation icons.
- **Totals & Subtotals:** Activate row subtotals, column subtotals, and overall Grand Totals.
- **Layout Formatting:** Apply Stepped layout and style presets for corporate financial standards.

SECTION 8 — SLICERS & FILTERING ARCHITECTURE

Implement a robust filtering ecosystem across the report canvas:

- **Interactive Slicers:** Deploy responsive slicers for **Year** , **Quarter** , **Month** , **Region** , **Category** , **Product** , **Customer Segment** , **Employee** , and **City** (utilizing dropdown and tiles layout).
- **Filter Pane Governance:**
 - **Visual Level Filter:** Applied to isolate outlier points on specific charts.
 - **Page Level Filter:** Confine specific analytical scopes per canvas page.
 - **Report Level Filter:** Establish universal global baseline criteria across all report tabs.
 - **Relative Date Filter:** Dynamically restrict operational views to "Last 30 Days" / "This Calendar Year".
 - **Top N Filter:** Integrated within visualization filters for rank restriction.

SECTION 9 — TOP N FILTERING APPLICATION

Configure dynamic Top/Bottom boundaries utilizing both Filter Pane parameters and DAX formulas:

- **Top 5 Products:** Restrict visual display to isolate the Top 5 revenue-generating products based on **Total Sales**.
- **Top 3 Regions:** Highlight the Top 3 most profitable territories evaluated by **Total Profit**.
- **Bottom 5 Products:** Filter to isolate the 5 lowest performing product lines by net profit for mitigation analysis.

SECTION 10 — CONDITIONAL FORMATTING RULES

Implement conditional formatting rules across the Matrix and Table visuals:

- **Font / Background Color:** Apply background tinting — **Profit > 0** (Soft Green) vs **Profit < 0** (Soft Red).
- **KPI Icon Indicators:**
 - **Profit Margin % > 30%** → Green Circle / Upward Arrow.
 - **Sales < Sales Target** → Red Flag / Warning Indicator.
- **Data Bars:** Embed in-cell gradient **Data Bars** on the **Sales** metric column for visual scanning.
- **Continuous Color Scale:** Apply 3-color gradient formatting (Min – Mid – Max) on the **Profit** measure.

SECTION 11 — CUSTOM REPORT PAGE TOOLTIPS

Design a modern, dedicated micro-canvas Tooltip page (Page Type: Tooltip):

- **Page Configuration:** Set page size to **Tooltip** (320 × 240 px) and enable "Allow use as tooltip".
- **Content Display:** When hovering over any product on visual charts, the tooltip must dynamically present: **Product Name**, **Total Sales**, **Total Profit**, **Quantity Sold**, **Profit Margin %**, and current **Sales Rank**.
- **Binding:** Bind the custom tooltip page to all relevant primary charts in Section 6.

SECTION 12 — DRILL-THROUGH ACTION CONFIGURATION

Create a granular target detail page named **Product Detail Page**:

- **Drill-Through Target Fields:** Add **Products[Product Name]** to the drill-through filter well.
- **Target Page Visuals:** Integrate historical **Sales Trend** line chart, **Profit Trend**, filtered **Customer Transaction List**, and **Region-wise Sales Distribution**.
- **Navigation:** Verify automatic creation and functionality of the "Back" navigation button.

SECTION 13 — BOOKMARKS & NAVIGATION BUTTONS

Implement bookmark states and interactive button triggers for intuitive user navigation:

- **Bookmark 1 — "Executive Dashboard":** Preserves visual view of top-level KPI cards and macroeconomic trends.
- **Bookmark 2 — "Sales Analysis Dashboard":** Displays expanded matrix tables, category slicers, and regional charts.
- **Interactive Navigation Buttons:**
 - **Home Button**: Returns user to page 1.
 - **Sales Analysis Button**: Triggers Bookmark 2 transition.
 - **Reset Filters Button**: Clears active slicers and restores defaults using bookmark actions.

SECTION 14 — MULTI-PAGE DYNAMIC DASHBOARD STRUCTURE

Structure the Power BI report file into four cohesive, professionally formatted canvas pages:

Page Name	Core Dashboard Components & Layout
Page 1 — Executive Overview	Executive KPI cards bar, primary KPI trend visual, monthly sales trend line, high-level slicers.
Page 2 — Sales Analysis	Region breakdown, Product Category matrix, Segment donut chart, combo target comparison.
Page 3 — Customer Analysis	Customer Segment demographics, City-wise geographic map/table, customer order distribution.
Page 4 — Employee Performance	Representative sales performance rankings, profit contribution by employee, incentive variance.

SECTION 15 — BONUS ADVANCED DAX COMPETENCIES

Demonstrate mastery over advanced context transition, filter manipulation, and variables:

CALCULATE()	FILTER()	ALL()	ALLSELECTED()	KEEPFILTERS()	RANKX()	TOPN()	DATESYTD()
DATEADD()	SAMEPERIODLASTYEAR()	TOTALMTD()	TOTALQTD()	TOTALYTD()	SWITCH()	VAR ... RETURN	

Requirement: Formulate composite measures applying **VAR / RETURN** construct to eliminate redundant calculations, optimize DAX query plan engine performance, and utilize **ALLSELECTED** for accurate denominator calculation in percentage-of-total metrics.

FINAL PRACTICAL ASSESSMENT CHECKLIST

- ☐ Data Modeling (Star Schema 1:*)
☐ Autonomous Date Dimension Table
☐ Power Query Transformations & ETL
☐ 15+ Advanced DAX Measures
☐ Calculated Columns with IF / SWITCH
☐ Executive KPI Cards & KPI Visuals
☐ MoM, YoY & YTD Time Intelligence
☐ Advanced Visuals (Waterfall, Ribbon, Combo)
- ☐ Multi-level Matrix with Drill Down
☐ Interactive Slicers & Filter Scopes
☐ Top N & Bottom N Ranking Logic
☐ Conditional Formatting (Icons, Bars, Scales)
☐ Dynamic Report Page Tooltip
☐ Dedicated Drill-Through Target Page
☐ Bookmarks, Navigation & Reset Filters
☐ 4-Page Integrated Dashboard Design

— • END OF PRACTICAL EXAMINATION PAPER • —